

Simple regression and OLS

Slide route: Lecture3v2.tex: regression line through large-sample distribution. Stock and Watson Chapter 4.

Session 4 | A line, an estimator, an assumption

Outcomes: Compute an OLS line, distinguish errors from residuals, interpret slope units, and state the role of conditional mean zero.

Opening A line can summarize an association. What extra argument would let us call its slope a causal effect? Keep that question on the board throughout the algebra.

100-minute route

0–10	Retrieve covariance and conditional expectation.
10–25	Population model, fitted line, errors versus residuals.
25–45	OLS objective and the three-observation example.
45–55	Interpretation and pause.
55–70	R^2 , SER, fit versus causality.
70–90	Causal assumptions; unbiasedness and sampling variation.
90–100	Exit ticket; hand off inference to Lecture 4.

Board example | Compute, do not memorize

Use with slides Lecture3v2 slides 4–9: The regression line, Population Linear Regression Model, and OLS Estimator. Begin the three-point calculation when slide 7 introduces OLS; use slides 8–9 for the first-order conditions and residual identities.

Use synthetic observations $(x, y) = (1, 2), (2, 3), (3, 5)$.

$$\bar{x} = 2, \quad \bar{y} = 10/3, \quad \hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{3}{2} = 1.5.$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 1/3.$$

Predictions are $11/6, 10/3, 29/6$; residuals are $1/6, -1/3, 1/6$. Their sum is zero, and $\sum x_i \hat{u}_i = 0$.

Derivation cue Differentiate $\sum (y_i - b_0 - b_1 x_i)^2$ with respect to both parameters. The first equation puts the fitted line through $(\bar{x}, \bar{y})$; substitute it into the second to obtain the slope formula.

Distinction Population u_i is unobserved. Sample $\hat{u}_i = y_i - \hat{y}_i$ is computed. OLS residual orthogonality is an algebraic identity, not evidence of exogeneity.

Use the same example | Fit and interpretation

Use with slides Lecture3v2 slides 11–15: Interpretation, Goodness of Fit, R^2 , SER, and Goodness of Fit again. Use the same three observations to compute R^2 on slide 13 and SER on slide 14.

$$SSR = \sum \hat{u}_i^2 = 1/6, \quad TSS = \sum (y_i - \bar{y})^2 = 14/3.$$

Here SSR means residual sum of squares; always define the notation because some texts use it for regression sum of squares.

$$R^2 = 1 - \frac{1/6}{14/3} = \frac{27}{28} \simeq .9643, \quad SER = \sqrt{\frac{1/6}{3-2}} = \sqrt{1/6} \simeq .408.$$

These three observations demonstrate arithmetic, not reliable asymptotic inference.

Units exercise If y is a test score and x is class size, slope 1.5 means an associated 1.5-point difference per additional student. If x is divided by 10, the numerical slope multiplies by 10. Predictions do not change.

Causality and sampling

Use with slides Lecture3v2 slides 16–26: conditions for causality, conditional mean zero, i.i.d. draws, outliers, and OLS sample properties. Use the bias expression beside slides 18–19 before moving to large-sample distribution.

For $Y_i = \beta_0 + \beta_1 X_i + u_i$, a zero conditional mean $\mathbb{E}(u_i | X_i) = 0$ says omitted determinants average to zero at each regressor value. Richer districts may have both different class sizes and different resources, violating this condition.

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (X_i - \bar{X}) u_i}{\sum_i (X_i - \bar{X})^2}.$$

Use the expression to explain why variation in X is needed and why correlation between X and omitted determinants is a problem. Discuss random sampling and finite-moment conditions before invoking large-sample normality.

Side question Does a high R^2 establish causality? No. Does a low R^2 rule out a causal effect? No. Fit and identification answer different questions.

Exit answers Residuals sum to zero when an intercept is included. That fact cannot verify $\mathbb{E}(u | X) = 0$. The intercept may have no sensible substantive interpretation when $X = 0$ is outside the observed range.

If short / preparation Assign the detailed asymptotic variance algebra for review. Preserve the numerical OLS calculation and exogeneity discussion. Read Chapter 5 before session 5.

Source: Lecture3v2.tex; synthetic three-observation calculation.